The Selling closure

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Abstract

Selling reduction does not return one cell. For a lattice with a right angle it returns several obtuse superbases, and Kurlin (2026) showed that their number is finite — between one and sixteen — and gave them for each lattice type. This paper calls the set of obtuse superbases, together with the integer changes of basis between them, the Selling closure, and makes three points about it. First, the closure is a network of reduced cells joined by integer matrices, and the lattice symmetries are the steps in that network that leave the metric tensor unchanged. In a primitive monoclinic lattice two of the four symmetries relabel the reduced cell and two carry it to a different reduced cell; in a primitive cubic lattice only 12 of the 48 symmetries can be seen in one reduced cell. The network depends only on which conorms vanish and which are equal, so there is one network per Delone sort, and the 24 sorts are recovered here from the six possible patterns of vanishing conorms, with the point groups as output. Second, a measured cell lies near a right angle rather than on it, and its exact closure jumps with the noise. The repair is to idealise the cell onto its nearest pattern of

zeros, take the exact closure there, and keep the members that fit the measured cell within a band; this is proved to return exactly the cells that Selling reduction would accept with its boundary moved from zero to $-\tau$, provided the shortest lattice vector is long compared with τ. The obvious alternative — walking from the measured cell across any conorm within the band — misses members. Third, reindexing a new cell onto a reference reads one step of the reference's network; from a single reduced cell of the new cell the true operator is found, with a stated certificate, and a cache built without the idealisation step can lose it. Every quantitative statement is tied to a named computational check.

1. Introduction

A lattice has infinitely many unit cells, and reduction picks one. That works until two cells are equally good, which happens at every right angle: there the rule has to choose, and its choice jumps while the lattice itself changes smoothly. Niggli reduction, Selling reduction and the S^6 methods of Andrews and Bernstein all meet this boundary, and each handles it in its own way (Niggli, 1928; Delone, 1932; Andrews *et al.*, 2019; Andrews & Bernstein, 2023).

Selling reduction offers a cleaner way out, because its answer at a right angle is not ambiguous but multiple. A lattice on a right angle has several obtuse superbases, Kurlin (2026) proved that there are never more than sixteen, and they can all be written down. This paper takes the whole set as the answer. Together with the integer matrices that convert one member into another, the set is called here the *Selling closure*. The name is new; Kurlin's results concern the set of obtuse superbases and do not use it.

The paper is organised around one monoclinic cell, introduced in Section 2, and it does three things with the closure.

1. It shows that the closure is a small network — reduced cells as nodes, integer matrices as steps — in which the lattice symmetries appear as the steps that leave the metric tensor unchanged. Some symmetries return a reduced cell to itself and can be seen in that cell; the others carry it to a different reduced cell and cannot. There is one such network per Delone sort, and the 24 sorts come out of the construction rather than going into it (Section 4.1).

2. It handles measured cells, which lie near right angles rather than on them. The method is to idealise the measured cell onto the nearest pattern of exact zeros, take the exact closure there, and keep what fits the measurement within a band. This is proved to give exactly the right set, and the obvious alternative is shown to miss members (Section 4.3).

3. It shows that reindexing a new cell onto a reference is reading one step of the reference's network, and that one reduced cell of the new cell is enough, with a certificate that the true operator has been found (Section 4.4).

The mathematics is confined to Appendix A. Every number in the text carries a mark [*znn*] pointing to the computational check that reproduces it (Appendix C).

2. The Selling closure, by example

2.1. Superbases and conorms

A Selling reduction does not work with three cell vectors but with four: the cell vectors a, b, c and their negative sum $d = -a - b - c$. The four together are a *superbase*; any three of them form a basis of the lattice. The six dot products between pairs, taken with a minus sign,

$$p_{ab} = -a \cdot b, \quad p_{ac} = -a \cdot c, \quad p_{bc} = -b \cdot c, \quad p_{ad} = -a \cdot d, \quad p_{bd} = -b \cdot d, \quad p_{cd} = -c \cdot d,$$

are the *conorms* (Selling, 1874; Conway & Sloane, 1992). The superbase is *obtuse*, which is to say Selling-reduced, when all six conorms are non-negative: every pair of the four vectors meets at 90° or more. The metric tensor of the cell is a linear combination of the conorms, so the six conorms are an alternative set of six cell parameters.

Selling reduction is a repeated *flip*. If a conorm is negative, say $p_{ab} < 0$, replace a by $-a$ and add the old a to the two remaining vectors: the new superbase is $(-a,\ b,\ c+a,\ d+a)$. The flipped conorm changes sign and the total $p_{ab} + \cdots + p_{cd}$ decreases, so the procedure stops, and it stops at an obtuse superbase. The same flip applied at a conorm that is exactly zero leads from one obtuse superbase to another obtuse superbase of the same lattice. This is the whole reason the reduced cell is not unique.

2.2. A primitive monoclinic cell

Take $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$. Its reduced superbase is $M_0 = (d, a, b, c)$ with $d = -a - b - c$, and its conorms are

$$p_{ab} = 0, \quad p_{bc} = 0, \quad p_{ac} = 260.5, \quad p_{ad} = 639.5, \quad p_{bd} = 1600.0, \quad p_{cd} = 2239.5 \text{ Å}^2.$$

The two zeros are the right angles α and γ at the twofold axis b, and every primitive monoclinic lattice has them. No other conorm is small, so the lattice is not near any higher symmetry; the next conorm is 260.5 Å^2 away.[z46]

Because two conorms are exactly zero, two flips lead to other obtuse superbases, and from those, two more. The closure has six members (Table 1). Each has two zero conorms, each flip leads to a neighbour, and the six form a ring, M_0–M_1–M_5–M_4–M_3–M_2–M_0. All six carry the same six conorm values, 0, 0, 260.5, 639.5, 1600.0 and 2239.5 Å^2, in different positions. They are different cells: in its own basis each member has, in general, a different metric tensor.

Table 1. *The Selling closure of the monoclinic cell $a = 30$, $b = 40$, $c = 50$ Å, $\beta = 100°$: six obtuse superbases, each given up to the order and the common sign of its four vectors, and the flips at zero conorms that connect them. Every member carries the same six conorm values in a different arrangement.*

member	four vectors	zero-conorm flips lead to
M_0	$-a-b-c,\ a,\ b,\ c$	M_1, M_2
M_1	$-b-c,\ -a,\ b,\ a+c$	M_0, M_5
M_2	$-a-c,\ a+b,\ -b,\ c$	M_0, M_3
M_3	$a+c,\ b-c,\ -b,\ -a$	M_2, M_4
M_4	$a-b+c,\ -c,\ b,\ -a$	M_3, M_5
M_5	$a-b,\ c,\ b,\ -a-c$	M_1, M_4

Any two members are related by an integer matrix with determinant ± 1, a change of basis. These matrices are the *steps* of the closure. Composing the steps around the ring is consistent, as it must be: a step from M_0 to M_1 followed by one from M_1 to M_5 is the step from M_0 to M_5.

2.3. Where the symmetry is

The lattice has point group $2/m$: the identity, the inversion, the twofold along b and the mirror normal to it. Each is an integer matrix that leaves the metric tensor unchanged, and each therefore maps the reduced superbase M_0 onto some obtuse superbase, that is, onto some member of the closure. Table 2 shows where.[z46]

Table 2. *The four symmetries of the monoclinic lattice of Table 1, applied to the reduced cell M_0. Matrices act on coordinates in the basis a, b, c.*

symmetry	matrix	M_0 is carried to	visible in M_0?
1	$\mathrm{diag}(1,1,1)$	M_0	yes, a relabelling of M_0
$\bar{1}$	$\mathrm{diag}(-1,-1,-1)$	M_0	yes, a relabelling of M_0
2 along b	$\mathrm{diag}(-1,1,-1)$	M_4	no, a different reduced cell
m normal to b	$\mathrm{diag}(1,-1,1)$	M_4	no, a different reduced cell

The identity and the inversion carry M_0 to itself; they only reorder or negate its four vectors, and anyone looking at the reduced cell can see them. The twofold does something else. It sends $a \mapsto -a$, $b \mapsto b$, $c \mapsto -c$, so it carries $(-a-b-c,\ a,\ b,\ c)$ to $(a-b+c,\ -a,\ b,\ -c)$, and that is not a reordering of M_0: it is the member M_4, three flips around the ring. Read in the twofold's order, $(-a, b, -c)$, the basis of M_4 has exactly the metric tensor of (a, b, c). Read in the order of Table 1, $(-c, b, -a)$, it

has a different one, with the conorm values 639.5 and 2239.5 exchanged.[z46] Any step from M_0 to M_4 changes the cell; the twofold and the mirror are the two steps that change it into a cell with the same metric tensor.

So, in this lattice, half of the point group is not visible in the reduced cell. It lives in the steps between reduced cells. Seen from M_4 the roles are the same: identity and inversion relabel M_4, and the twofold and mirror are steps back to M_0.

2.4. Terms used below

Two phrases recur. The *zero pattern* of a lattice is which of the six reduced conorms vanish; the monoclinic cell above has two zeros, in the slots *ab* and *bc*. Lattices with the same zero pattern have the same closure members, written as the same integer combinations of the reduced cell vectors, and the same steps between them; only the conorm values differ (this is the stratum proposition of the companion paper, Zwart, 2026). The *symmetry steps* of a lattice are the steps between members that leave the metric tensor unchanged; there is one out of each member for each element of the point group.

3. Methods

3.1. Computing the closure and its symmetry steps

For each of the 24 Delone sorts a generic lattice was built from the sort's Selling pattern (Andrews *et al.*, 2023), which says which conorms vanish and which are equal; the free values were drawn at random. Its closure was enumerated by flips at the exact zeros, starting from the reduced superbase, and its symmetries were found by testing every member, in every order and sign of its four vectors, against the metric tensor. The symmetries that carry the reduced superbase to itself were counted separately from those that carry it elsewhere.

3.2. Recovering the Delone sorts

Only six zero patterns are possible, up to relabelling, if the four vectors of the superbase are to stay connected by non-zero conorms. For each, a generic lattice with that zero pattern was built and its closure enumerated. Each member, in each order of its four vectors, rearranges the six conorm values; the rearrangements that keep the zeros in place form a finite set of permutations. A pattern of equal conorms is then a candidate sort, and it is a genuine sort when imposing those equalities on a generic lattice produces a lattice whose symmetries enforce exactly those equalities and no others. Patterns found from different members of one closure describe the same lattices and were counted once. Nothing about Bravais types or point groups was supplied.

3.3. Measured cells: idealise, enumerate, filter

A measured cell is first Selling-reduced to a superbase F_0. Its small conorms are then idealised: every conorm of F_0 that is at most 4τ is set to zero, all of them at once, where τ is a band in Å^2 set from the measurement error (Appendix B). This gives an idealised lattice, written Λ^*, with an exact zero pattern. Its exact closure is enumerated by flips at those zeros, keeping each member as integer combinations of the reduced cell vectors. Finally each member is evaluated with the *measured* metric tensor, and the members whose six conorms are all at least $-\tau$ are kept. The target of this procedure is the *widened closure*

$$\mathcal{S}_\tau(\mathbf{G}) = \big\{\, F \;:\; \text{all six conorms of } F \text{ under } \mathbf{G} \text{ are } \geq -\tau \,\big\}, \tag{1}$$

the set of superbases that Selling reduction would accept if its boundary were moved from 0 to $-\tau$. The *depth* of a superbase is the size of its most negative conorm, so a superbase is in the widened closure when its depth is at most τ.

3.4. Reindexing from one reduced cell

A reference cell R is stored as its widened closure, each member as integer combinations of R's reduced cell vectors. A new cell N is Selling-reduced once. Pairing N's reduced superbase with each stored member of R, in each order and sign of the four vectors, gives a candidate integer operator $\mathbf{P}$ with $\mathbf{G}_N \approx \mathbf{P}^\mathsf{T}\mathbf{G}_R\mathbf{P}$, and the candidates are ranked by the residual of that equation, measured in R's frame.

3.5. Verification

Each quantitative statement was checked by code. Closure memberships were compared with brute force: every superbase whose coordinates relative to F_0 lie in $[-2, 2]$ (135 408 bases of determinant ± 1) was tested against equation (1). Symmetry counts and sort derivations were done by exhaustive enumeration. Test lattices were specified by their conorms, so F_0 is obtuse by construction. The measured worked example throughout is the monoclinic cell 30.02, 39.97, 50.03 Å, 90.25°, 92.05°, 89.80°, which is the cell 30–40–50 Å, $\beta = 92°$ perturbed by 14.4 Å^2 in the Frobenius norm of the metric tensor. The checks are the test files listed in Appendix C.

4. Results

4.1. One network per Delone sort

What was seen in the monoclinic cell holds in general. The closure of any lattice is connected by flips at zero conorms, and every step between members — including every symmetry step — can be written as a sequence of such flips followed by a reordering of the four vectors (Appendix A, Lemma A5). A symmetry is thus never more than a short walk in the closure. Out of each member there are exactly as many symmetry steps as the point group has elements, and for every one of the 24 sorts that number came out as the order of the holohedry: 48 for cubic, 24 for hexagonal,

16 for tetragonal, 12 for rhombohedral, 8 for orthorhombic, 4 for monoclinic and 2 for triclinic (Table 3).[z46]

Table 3. *The Selling closure of each Delone sort, computed for a generic lattice of the sort's Selling pattern (Andrews* et al.*, 2023; the six symbols are the Selling parameters, equal letters meaning equal values,* 0 *meaning zero). Members are counted up to the order and sign of the four vectors. "Visible in F_0" is the number of symmetries that carry the reduced superbase to itself; the rest carry it to another member. "Members reached" is how many members the symmetries carry F_0 to. The last two columns depend on which member one reduces to; the values are for the pattern as written.*

sort	Bravais	pattern	zeros	members	|point group|	visible in F_0	reached
C1	cI	rrrrrr	0	1	48	48	1
C3	cF	rr0rr0	2	3	48	16	3
C5	cP	000rrr	3	16	48	12	4
H4	hP	00rrrs	2	6	24	4	6
T1	tI	rrsrrs	0	1	16	16	1
T2	tI	rr0rrs	1	2	16	8	2
T5	tP	000rrs	3	16	16	4	4
R1	hR	rrrsss	0	1	12	12	1
R3	hR	rr0sr0	2	3	12	4	3
O1A	oF	rrsrrt	0	1	8	8	1
O1B	oI	rstrst	0	1	8	8	1
O2	oI	rs0srt	1	2	8	4	2
O3	oI	rs0rs0	2	3	8	8	1
O4	oS	00rsst	2	6	8	4	2
O5	oP	000rst	3	16	8	2	4
M1A	mC	rrsttu	0	1	4	4	1
M1B	mC	rstrsu	0	1	4	4	1
M2A	mC	rs0stu	1	2	4	2	2
M2B	mC	rs0rst	1	2	4	4	1
M3	mC	rs0ts0	2	3	4	4	1
M4	mP	00rstu	2	6	4	2	2
A1	aP	rstuvw	0	1	2	2	1
A2	aP	rs0tuv	1	2	2	2	1
A3	aP	rs0tu0	2	3	2	2	1

Two different things fix the network. Which members exist, and which steps join them, is fixed by the zero pattern alone. Which of the steps are symmetries is fixed by the Bravais type. The two do not line up one to one. Primitive monoclinic, C-centred orthorhombic and primitive hexagonal lattices all have two zeros and six members, and differ only in how many of the steps are symmetries: 4, 8 and 24 out of each member. In the other direction, body-centred tetragonal lattices have no zero or one zero depending on c/a, and C-centred monoclinic lattices occupy five sorts with three different zero patterns, depending on the shape of the cell. Within one sort the shape

plays no part: two lattices of the same sort have the same members, the same steps and the same symmetry steps, and differ only in the numbers — conorm values, distances, prices — attached to that fixed structure.

The practical consequence is the last two columns of Table 3. In the sorts with zero conorms, the symmetries that can be seen in the reduced cell are often a small part of the point group: 2 of 8 for primitive orthorhombic, 4 of 16 for primitive tetragonal, 12 of 48 for primitive cubic, 4 of 24 for hexagonal, 4 of 12 for the rhombohedral sort R3.[z46] The rest carry the reduced cell to another member. Those are exactly the operators that symmetry assignment and reindexing have to find, and they are properties of the closure, not of any one cell.

4.2. The 24 Delone sorts fall out

The procedure of Section 3.2, which starts from six zero patterns and knows nothing of Bravais types, returns exactly 24 patterns of equal conorms, and they are the 24 Delone sorts as listed by Andrews *et al.* (2023), none missing and none extra.[z48] The point groups were not supplied; they come out as the holohedries above, and the member counts come out as Kurlin's 1, 2, 3, 6 and 16.[z48] The three patterns that Andrews *et al.* describe as degenerate boundaries (O3, M2B, M3) arise in the same way as the others. A Delone sort is, in these terms, a pattern of zero and equal conorms such that a lattice with that pattern has exactly the symmetries the pattern implies: the zeros fix the members and steps, the equalities fix which steps are symmetries.

4.3. Measured cells

A measured monoclinic cell has $\alpha = 90.25°$, not $90°$, and its reduced conorms p_{ab} and p_{bc} are small numbers of either sign rather than zeros. If they are positive, the flips that produced M_1 and M_2 in Table 1 now lead to superbases with slightly negative

conorms, which Selling reduction rejects, and the exact closure has one member instead of six. The twofold and the mirror, which were steps to M_4, are then steps to a cell that is no longer in the set. This is the jump the introduction warned of, and it is why the paper widens the closure.

The widened closure is a well defined object. Equation (1) mentions no starting cell, no order of operations and no frame, so the widened closure is the same whoever computes it. The cell at which reduction with a $-\tau$ boundary *stops* is not well defined: started from 400 random unreduced bases, with two different flip orders, reduction of the measured worked cell at $\tau = 43.2\,\text{Å}^2$ stopped at six different cells, and reduction of a noisy orthorhombic cell at sixteen. Every stopping cell was a member of the widened closure, but on a noisy cubic cell at $\tau = 10\,\text{Å}^2$ the 400 restarts found 15 of the 16 members.[z45] Restarting is not a way to enumerate the set; one has to compute the set.

What the widened closure means. A superbase belongs to the widened closure of a measured cell exactly when it is the exactly reduced superbase of some lattice that differs from the measured one by at most τ in each of that superbase's own six conorms (Proposition 1, Appendix A). One direction says that the exact members of every nearby lattice are kept; the other says that nothing else is, because clipping a member's negative conorms to zero produces the nearest lattice for which it is exactly reduced. The widened closure is thus the union of the exact closures of all lattices consistent with the measurement.

The procedure is complete. If the shortest lattice vector has squared length greater than 32τ, the idealise–enumerate–filter procedure of Section 3.3 returns exactly the widened closure (Theorem 2, Appendix A). The proof shows that any superbase within the band is an exact member of the idealised lattice Λ^*, so that enumerating Λ^*'s closure cannot miss it; it rests on two small integer constants, 4 and 8, found by

exhaustive enumeration.[z44] On every test lattice and band, including walls placed between τ and 4τ, the procedure returned the brute-force set.[z44] The condition on the shortest vector is sufficient rather than sharp: for the measured worked cell it allows τ up to 28.2 Å^2, and the procedure is still exact at 43.2 and at 60 Å^2.[z44] For a shortest cell edge of 30 Å the condition allows bands up to about 28 Å^2.

Idealise all small conorms together. Setting the small conorms to zero one subset at a time gains nothing: the closure of any partial idealisation is contained in the closure of the full one, and on the measured worked cell and a noisy cubic cell the union over all eight subsets is exactly the full idealisation's sixteen members.[z46] The threshold 4τ is what the proof needs. On dense grids of two-wall configurations, idealising at 2τ was already exact, and idealising at τ was not.[z46]

Why not simply walk from the measured cell. The natural alternative is to start from the measured reduced cell and flip across any conorm that is within some band of zero. Table 4 compares four versions of that walk with the procedure above on 80 test cells with walls placed at awkward positions. All four walks miss members. The cause is a member that can only be reached through a non-member: with one wall at 1.2τ and another at 0.5τ, the flip across the first produces a conorm of -1.2τ, outside the band, and only a second flip, across the other wall, lands on a member of depth 0.7τ. A walk that refuses to leave the band never takes the first flip. On the idealised lattice both walls are exactly zero and the two flips cost nothing. Idealising at τ instead of 4τ fails on the same configuration, returning six of seven members.[z44]

Table 4. *Four walks on the measured metric, and the idealise–enumerate–filter procedure, against brute force on 80 cells with walls at awkward positions (orthorhombic, cubic, plate, monoclinic and hexagonal zero patterns;* $\tau = 0.1$ *and* 0.5 *of* $\lambda_1/32$*).*

procedure	cells with a wrong set
flip wherever $\lvert p\rvert \le \tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \le \tau$, filter at the end	13
flip wherever $\lvert p\rvert \le 2\tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \le 2\tau$, filter at the end	2
idealise at 4τ, enumerate the exact closure, filter	0

Where the easily missed members are. The members that a walk misses have a recognisable form. Their most negative conorm is not one of the measured cell's small conorms but the *difference* of two of them: a near-equality in the measured cell that is a near-zero in theirs. They occur only where three walls meet, as at an orthorhombic or cubic corner, and only when one small conorm z is at most τ and the other, y, lies between τ and $z + \tau$; they never occur at the two-wall corner of a monoclinic or hexagonal cell.[z46] They carry no symmetry: a lattice lying exactly on such a difference-wall has point group $\bar{1}$.[z46] An easily missed member is simply another reduced cell within the band, and a second measurement of the same lattice can well reduce to it.

One enumeration serves every band. The idealised lattice changes only when 4τ crosses one of the measured cell's conorms, so one enumeration of its closure serves a whole range of bands, with each member labelled by its depth. The measured worked cell has reduced conorms $(49.5, 1588.9, 2444.7, 4.2, 847.5, 4.5)\,\text{Å}^2$. Idealised at $4 \times 43.2\,\text{Å}^2$, all three small conorms become zeros, the idealised lattice is orthorhombic with sixteen members, and their depths in the measured metric are 0, 4.2, 4.5, 4.5, 8.7, 8.7, 45.0, 49.5, 49.5, 53.7 (four times), 57.9, 62.5 and $62.5\,\text{Å}^2$. The widened closure therefore has six members for any band between 8.7 and $45.0\,\text{Å}^2$ — the six members of the unperturbed cell's closure, written with the same integer coefficients — seven at 45.1, nine at 50 and fourteen at $60\,\text{Å}^2$.[z45]

Pseudo-symmetry has a price and a threshold. The steps of the idealised lattice's closure are exact integer matrices, and its symmetries — here the eight of *mmm* — are among them. Applied to the measured metric tensor they are not exact; each changes it by a residual, which is the *price* of that pseudo-symmetry in Å^2. Members with the same depth enter the widened closure together, and the depth at which a member enters is the band at which the pseudo-symmetry it carries becomes visi-

ble: on the worked cell the members entering at $45.0\,\text{Å}^2$ are the ones that carry the near-orthorhombic twofolds. Within the band, the steps that join members, the steps through members just outside the band, and the priced symmetries together describe the measured cell completely; the structure is that of the idealised lattice, restricted to what the measurement can support.

4.4. Reindexing reads a step

Two settings of one lattice are related by one operator composed with any lattice symmetry, and every one of those operators is a step from a member of one closure to a member of the other. On the measured worked pair at $\tau = 3\eta = 43.2\,\text{Å}^2$, pairing the six members of the unperturbed cell with the six of the perturbed one gives twelve candidate operators at a residual of $14.4\,\text{Å}^2$ — four distinct operators, the $2/m$ coset, at a residual equal to the perturbation — and nothing else below $1378\,\text{Å}^2$. Once the band passes $45.0\,\text{Å}^2$ the members that carry the near-orthorhombic symmetry enter: at $\tau = 60\,\text{Å}^2$, fourteen members give twelve operators at 14.4, ten at 150.7 and nothing below $992\,\text{Å}^2$.[z39] The band decides which pseudo-symmetries are offered, and the closure says exactly which.

A new cell needs no closure of its own. Its single reduced superbase, paired with the stored widened closure of the reference, is enough. For fifteen references near orthorhombic, cubic, plate, monoclinic and hexagonal corners, each with twelve new cells carrying a random metric error of declared size and a random reindexing, the new cell's reduced superbase was always a member of the reference's widened closure, with the correct integer coefficients, and the true operator was always among the candidates: 180 of 180 cases, with the band set from the declared error as in Appendix B.[z47]

Completeness of the stored closure matters. A new cell was constructed whose only reduced superbase is one of the easily missed members of an orthorhombic corner,

and matched against two stored closures of the reference. The closure built by the procedure of Section 3.3 offers candidate operators at 7.83, 8.01 and 8.84 Å^2, each with its negative; the closure built by idealising only at τ offers the first two and has lost the third, which is $\pm\mathbf{I}$, the true operator.[z46] The best-scoring candidate is still found, through a pseudo-symmetry, but the set of operators reported is incomplete.

That the true operator is among the candidates can be guaranteed before matching. If the new cell's metric tensor, written in the reference frame, is within η of the reference's, and V is the sum of the squared lengths of the new cell's three reduced cell vectors, then the band $\tau = \sqrt{2}\,\eta\,V/(\lambda_{\min} - \eta)$, with $\lambda_{\min}$ the smallest eigenvalue of the reference metric tensor, suffices (Proposition 3, Appendix A). The guarantee held in all 180 cases, but it is conservative: its median value is 13η, well above the band that was actually needed.[z47]

5. Discussion

Mathematicians call a network like the closure — nodes, invertible steps between them, consistent composition — a *groupoid* (Brandt, 1927; Weinstein, 1996; Brown, 2006). The closure is the groupoid of integer changes of basis acting on the Selling-reduced cells of one lattice, and its symmetry steps form the sub-structure in which the metric tensor is preserved. Crystallography has used groupoids before, for the partial symmetries of order–disorder structures (*International Tables for Crystallography*, 2006, Vol. C, ch. 9.2), but, to my knowledge, not for reduced cells. Nothing new is needed from that theory; what is new is to notice that reduced cells form such a structure, that the Delone sorts are its 24 forms, that a measured cell inherits the structure of its idealised lattice, and that pseudo-symmetry is a priced step of it.

The network also names a difficulty the field has long worked around. A reduced cell, Niggli or Selling, shows only the part of the point group that relabels it, which

in the sorts with zero conorms is a small part (Table 3); the remainder are steps to other reduced cells. Tolerance-based symmetry assignment inspects one reduced cell's parameters, and the S^6 distances of Andrews *et al.* (2019) and Andrews & Bernstein (2023) reach the other cells through boundary transformations. The closure keeps all the cells with their integer relations, and the idealisation step makes that available for measured data with a statement of completeness.

What is proved should be kept apart from what is checked. Proved in Appendix A, from Kurlin's results and two integer constants obtained by exhaustive enumeration, are: that the widened closure is well defined and is the union of nearby exact closures; that idealise–enumerate–filter returns it when the shortest lattice vector exceeds 32τ in squared length; that the closure is connected by flips and reorderings; that partial idealisations add nothing; and the reindexing guarantee. Checked exhaustively but on generic lattices with random values, and therefore computations rather than proofs, are the identification of the 24 sorts and the counts of Table 3; a version in exact integer arithmetic would close that gap. Checked but not proved are three statements: that idealising at 2τ already suffices, that the true operator is found at the band set from the error rather than only at the conservative guarantee, and that the operators accepted within a band are exactly one operator composed with the idealised lattice's symmetries priced within that band. The first two held in every case tested.

The frame of reference enters in one place only. Everything about the widened closure is stated in terms of τ, a band on conorms, and needs no frame. Converting a measurement error on the metric tensor into τ does need one, because the Frobenius norm of an error depends on the basis it is written in. Appendix B gives the exact relation between a metric error and the resulting shift of each conorm, the worst-case factors by zero pattern (up to $2\sqrt{3}$ in some cubic frames, so that the customary $\tau = 3\eta$ is not a worst-case guarantee everywhere), a frame-free bound, and the route to take

when an error covariance for the metric tensor is available.

6. Conclusions

Selling reduction returns a small network of reduced cells, joined by integer matrices, in which the lattice symmetries are the steps that preserve the metric tensor. A reduced cell shows only the symmetries that return it to itself; in a primitive monoclinic lattice that is two of four, in a primitive cubic lattice twelve of 48. Which cells exist and how they are joined depends only on the pattern of zero conorms, which steps are symmetries only on the Bravais type, and lattices of one Delone sort share one network; the 24 sorts are recovered from six zero patterns with the point groups as output. A measured cell is idealised onto its nearest zero pattern at 4τ, the exact closure is enumerated there, and the members that fit the measured metric within τ are kept; this is exactly Selling reduction with its boundary at $-\tau$, it is unique, it is the union of the exact closures of all lattices within the band, and it is complete when the shortest lattice vector is long compared with τ. Walking from the measured cell instead misses members. Reindexing a new cell onto a reference reads one step of this network from a single reduced cell of the new cell, and completeness of the stored network is what keeps the true operator among the candidates.

Appendix A
Proofs

Terms. Write a lattice vector in any basis, $x = n_1 v_1 + n_2 v_2 + n_3 v_3$ with integer n_i, and record whether each coordinate is even or odd. This sorts the lattice into eight *parity classes*; two vectors are in the same class exactly when their difference is twice a lattice vector, so the classes do not depend on the basis (the classes of $\Lambda/2\Lambda$).

One class is the doubled vectors; the other seven are used here. A *partial sum* of a superbase is the sum of some of its four vectors; since the four add to zero, there are seven partial sums up to sign, one in each of the seven classes, and for the monoclinic superbase $(-a-b-c, a, b, c)$ they are a, b, c, $a+b$, $a+c$, $b+c$ and $a+b+c$. The fact used (Conway & Sloane, 1992; Kurlin, 2026, Lemmas 2.4, 2.7 and A.1) is that a superbase is obtuse exactly when each of its partial sums is a shortest vector of its parity class. The *excess* $e_c(F) = |v_c(F)|^2 - \mu_c$ of a superbase F in class c is the squared length of its partial sum $v_c(F)$ in that class minus the squared length μ_c of the shortest vector of the class; an obtuse superbase has zero excess in every class. A partial sum is written $w = \sum s_i v_i$ with $s \in \{0,1\}^4$ marking which vectors it adds; a vector of the same parity class has coordinates $s + 2k$ with k integer. Every lattice vector $x = \sum c_i v_i$ satisfies Selling's identity

$$|x|^2 = \sum_{i<j} p_{ij}\,(c_i - c_j)^2, \tag{2}$$

whatever the signs of the conorms, because the four vectors sum to zero. For a superbase F and two metrics, $\delta_F(\mathbf{G}, \mathbf{G}')$ is the largest change among F's six conorms.

Proposition 1. $\mathcal{S}_\tau(\mathbf{G}) = \{F : F \in \mathcal{G}(\mathbf{G}') \text{ for some } \mathbf{G}' \text{ with } \delta_F(\mathbf{G}, \mathbf{G}') \le \tau\}$. *Proof.* ($\supseteq$) If F is obtuse for $\mathbf{G}'$ and $\delta_F(\mathbf{G}, \mathbf{G}') \le \tau$, every conorm of F under $\mathbf{G}$ is at least $-\tau$. ($\subseteq$) If $F \in \mathcal{S}_\tau(\mathbf{G})$, the metric whose conorms in the frame of F are $\max(p_{ij}, 0)$ exceeds $\mathbf{G}$ by a positive semidefinite term, so it is positive definite; F is obtuse for it, at δ_F equal to the depth of F. Each negative conorm must move at least to zero, so no nearer metric makes F obtuse. □

Lemma A1 (class excess identity). Let F be any superbase, $w = \sum s_i v_i$ a partial sum and w' a vector of its parity class with coordinates $s + 2k$. Then $|w|^2 - |w'|^2 = -4\sum p_{ij}\gamma_{ij}$, with $\gamma_{ij} = \Delta k(\Delta k + d)$, $\Delta k = k_i - k_j$, $d = s_i - s_j$; each γ_{ij} is a non-negative integer, zero exactly when $\Delta k \in \{0, -d\}$. *Proof.* Equation (2), $(d + 2\Delta k)^2 - d^2 = 4\Delta k(\Delta k + d)$, and $t(t+d) \ge 0$ for integers t and $d \in \{-1, 0, 1\}$.

Lemma A2 (class gap). Let Λ^* be obtained from F_0 by setting its conorms $\le \theta$ to zero, be positive definite, and let m^* be the smallest conorm of F_0 above θ. A superbase not obtuse for Λ^* has $e_c \ge 4m^*$ in some class. *Proof.* In the frame of F_0, with w the partial sum of F_0 in class c — a shortest vector of its class for $\mathbf{G}$ and for Λ^* — Lemma A1 gives $e_c(F) = 4\sum p^0_{ij}\gamma_{ij}$, a sum of non-negative terms, and the excess for Λ^* is the same sum over the slots above θ. If those terms vanish in all seven classes, every partial sum of F is a shortest vector of its class for Λ^* and F is obtuse for it; otherwise some $\gamma_{ij} \ge 1$ multiplies a $p^0_{ij} \ge m^*$.

Lemma A3 (small excess in the band). If $\lambda_1 > 32\tau$ and every conorm of F is at least $-\tau$, then $e_c(F) \le 16\tau$ in every class. *Proof.* Call a slot large if $p_{ij} > 8\tau$, and let N be the negative slots. The large slots connect the four vectors, since otherwise a split crossed only by slots with $p_{ij} \le 8\tau$ has a partial sum of squared length at most $32\tau < \lambda_1$. With w' a shortest vector of the class, Lemma A1 in the frame of F gives $0 \le e_c(F) = 4\sum_N |p_{ij}|\gamma_{ij} - 4\sum_{p_{ij}>0} p_{ij}\gamma_{ij}$, so $\sum_{\text{large}} p\gamma \le \tau\sum_N \gamma$. If a large γ were positive, Lemma A4(ii) would give $8\tau\sum_{\text{large}}\gamma < \tau\sum_N\gamma \le 8\tau\sum_{\text{large}}\gamma$; so the large γ vanish, and Lemma A4(i) gives $e_c \le 4\tau\cdot 4$.

Lemma A4 (two constants on four points). Let L be a set of slots connecting the four vectors and N disjoint from L. (i) If $\gamma_{ij} = 0$ on L, then $\sum_N \gamma_{ij} \le 4$. (ii) $\sum_N \gamma_{ij} \le 8\sum_L \gamma_{ij}$. Both are attained. *Proof.* In (i) $|\Delta k| \le 1$ on L, so with $k_0 = 0$ all $|k_i| \le 3$, and the maximum is a finite enumeration over the 38 connected slot sets, the sets N and the 14 partial sums. In (ii) the ratio is enumerated for $|k_i| \le 15$; beyond, $Q = \sum_L \Delta k^2 > 75$, and $\sum_N \gamma \le \rho Q + \sqrt{|N|\rho Q}$, $\sum_L \gamma \ge Q - \sqrt{|L|Q}$, with $\rho \le 3+2\sqrt{2}$ the largest generalised eigenvalue of $\sum_N \Delta k^2$ against $\sum_L \Delta k^2$, keep it below 8 for every pattern. Equality in (ii): $L = \{01, 02, 13\}$, $N = \{03, 12, 23\}$, $s = (0, 0, 1, 1)$.[z44]

Theorem 2. Let F_0 be obtuse for $\mathbf{G}$, $\lambda_1 > 32\tau$, and Λ^* obtained by setting every conorm of F_0 that is at most 4τ to zero. Then Λ^* is positive definite and $\mathcal{S}_\tau(\mathbf{G}) =$

$\{F \in \mathcal{G}(\Lambda^*) : p_{ij}(F;\mathbf{G}) \geq -\tau\}$. *Proof.* If the slots of F_0 above 4τ did not connect the four vectors, some partial sum would have squared length at most $16\tau < \lambda_1$. The inclusion $\supseteq$ is the definition. For $F \in \mathcal{S}_\tau(\mathbf{G})$, Lemma A3 gives $e_c(F) \leq 16\tau$ in every class; the smallest surviving conorm exceeds 4τ, so Lemma A2 makes F obtuse for Λ^*. □

Lemma A5 (generation). Every step between members of the closure, and every symmetry, is a sequence of flips at zero conorms followed by a reordering and sign change of the four vectors. *Proof.* The members are reached from F_0 by flips at zero conorms (Kurlin, 2026; reproduced by every enumeration here), so these flips with the reorderings connect all members. A symmetry maps F_0 to an obtuse superbase, which is a member; as a matrix it is the flip path from F_0 to that member followed by a reordering. In the language of groupoids, the symmetry steps form the action groupoid of Aut on the members: its connected components are the orbits of members under Aut, and the steps from a member to itself are its stabiliser (Brown, 2006).

Lemma A6 (nesting). If $Z' \subseteq Z$ are sets of conorms of F_0 set to zero, every member of the closure for Z' is a member of the closure for Z. *Proof.* The conorms of every member of a closure are non-negative integer combinations of the free conorms of its zero pattern (the stratum proposition of Zwart, 2026; checked in [z46]); setting further free conorms to zero keeps them non-negative.

Lemma A7 (sorts as closed patterns of equalities). Fix a zero pattern Z and let Π_Z be the permutations of the six conorm slots produced by the members of the closure under all reorderings that keep the zeros in place. A lattice with zero pattern Z and a given pattern P of equal conorms has exactly the symmetries in Π_Z that preserve each class of P. *Proof.* A symmetry maps F_0 to a member with the same conorm vector, so it is an element of Π_Z that fixes the values; the members are complete by Kurlin's theorem, so no symmetry is missed. A permutation fixes a vector of values

exactly when it preserves each class of equal values. Hence the pattern of equalities of a generic lattice of a sort is the orbit partition of its symmetry group, and every such closed pattern is the pattern of the generic lattices of one sort.

Proposition 3 (reindexing guarantee). Let $\mathbf{G}'_N$ be the new cell's metric written in the reference frame, $\|\mathbf{G}'_N - \mathbf{G}_R\|_F \le \eta < \lambda_{\min}(\mathbf{G}_R)$, let F_N be its reduced superbase with basis vectors of summed squared length V, and let $\tau \ge \sqrt{2}\,\eta\,V/(\lambda_{\min}(\mathbf{G}_R) - \eta)$ with $\lambda_1(\mathbf{G}_R) > 32\tau$. Then F_N, with its true integer coefficients in the reference frame, is a member of $\mathcal{S}_\tau(\mathbf{G}_R)$, the procedure of Section 3.3 produces it, and pairing it with the new cell's reduced superbase proposes the true operator. *Proof.* Let u_i be the coordinates of F_N's basis vectors in the reference frame and $\Delta = \mathbf{G}'_N - \mathbf{G}_R$. Then $u_i^{\mathsf{T}}\mathbf{G}'_N u_i \ge (\lambda_{\min}(\mathbf{G}_R) - \eta)|u_i|^2$, so $\sum |u_i|^2 \le V/(\lambda_{\min} - \eta)$. In F_N's own frame the error is $U\Delta U^{\mathsf{T}}$, of Frobenius norm at most $\sum |u_i|^2\,\eta$, and in its own frame every conorm shift is at most $\sqrt{2}$ times that norm (Appendix B). The conorms of F_N under $\mathbf{G}'_N$ are non-negative, so under $\mathbf{G}_R$ they are at least $-\tau$, and Theorem 2 applies. The proposed operator for the pair is the change of basis between the two copies of F_N, which is the true one. □

Appendix B
From a metric error to a band

A conorm is linear in the metric, so a metric error $\Delta\mathbf{G}$ shifts the conorm of a member with coordinate vectors u_i, u_j by exactly $-u_i^{\mathsf{T}}\Delta\mathbf{G}\,u_j$; by Cauchy–Schwarz in the Frobenius product the shift is at most $c_{ij}\eta$, with $c_{ij} = \|\operatorname{sym}(u_i u_j^{\mathsf{T}})\|_F = [\frac{1}{2}(|u_i|^2|u_j|^2 + (u_i \cdot u_j)^2)]^{1/2}$, and an error along $\operatorname{sym}(u_i u_j^{\mathsf{T}})$ attains it. Every exact member of a stored closure survives in the widened closure of the measurement exactly when τ is at least the coverage bound $\tau_{\mathrm{cov}} = \max(c_{ij}\eta - p_{ij})$, read off the stored closure.[z36] Member vectors are not confined to $\{-1, 0, 1\}$; on two and three walls

they carry entries up to 3. On walls the largest constant is $\sqrt{2}$ with one zero, $\sqrt{5}$ with two, and with three zeros $2\sqrt{3}$ in six of the sixteen labelled cuboid frames — those whose zeros form a path with v_0 at an interior vertex — and at most $\sqrt{5}$ otherwise; on positive slots it is at most 7.2. In a $2\sqrt{3}$ frame an adversarial error of norm η drives a member to $-3.46\,\eta$, so $\tau = 3\eta$ is not a worst-case guarantee in every frame.[z36] In a member's own frame the member is the standard superbase, whose largest constant is $\sqrt{2}$, so $\tau = \sqrt{2}\,\eta^*$ with η^* the largest error norm over the members' own frames covers every member; it is uniform but loose, 2.8 times the reduced-frame η in the median case and up to 17 times.[z36] With a covariance Σ for the metric, the variance of each member conorm is the quadratic form of Σ on $\mathrm{sym}(u_i u_j^{\mathsf{T}})$, exact in any frame, and the band can be set slot by slot.

Appendix C
Computational checks

Each mark [z*nn*] refers to `manuscript/appendix_z/test_znn_*.py` in the `agentsg` repository.

mark	what is checked
z36	shift identity; constants by zero pattern and frame; adversarial loss at 3η; frame-free bound
z39	measured worked pair: widened closure by definition; candidate operators at $\tau =$ 43.2 and 60
z44	constants 4 and 8; class gap and excess; Theorem 2 against brute force; grey zone; worked cell
z45	reduction endpoints and restarts; walks on the measured metric; depths of the sixteen members
z46	the monoclinic example; closure of the 24 sorts; coset completeness; easily missed members; 2τ; nesting
z47	reindexing under a declared error: membership, true operator, guarantee
z48	the 24 Delone sorts derived from six zero patterns

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Synopsis

A lattice with a right angle has several Selling-reduced cells, not one. Kept together with the integer matrices that relate them, they form a small network in which every lattice symmetry is a step from one reduced cell to another (or back to itself); there is one such network per Delone sort. For a measured cell the network is recovered by idealising the cell onto its nearest zero pattern, and reindexing is a matter of reading one step of it.